Human Computer Interaction (HCI) – COSC6340 (3A) (For Preparing the Term Project)

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Objectives

- Affective Computing
- Sentiment Analysis Task
- Machine Learning and Categories
- Supervised Learning
  - Supervised Learning Workflow
  - Supervised Learning Data Set and Distribution
  - Supervised Learning algorithms
  - Evaluation Metrics

Affective Computing

Definition

- Affective computing is a filed that focuses on enabling machines to recognize, interpret, process, and simulate human emotions.
- It combines techniques from Artificial Intelligence (AI), Natural Language Processing (NLP), Computer Vision (CV), and Psychology to create systems that can understand and respond to human emotions effectively.

Sentiment Analysis

Definition

- Sentiment analysis in NLP is the process of using natural language processing (NLP) techniques to determine the emotional tone or sentiment expressed in a piece of text.
- It involves classifying text into predefined categories, such as positive, negative, or neutral, and can be extended to detect emotions like joy, anger, or sadness.

Emotion Recognition as a Classification Task (Sentiment Analysis)

Let

- $\mathbf{X} = \{\omega_2, \omega_2, ..., \omega_n\}$ be a sequence of words (tokens) in a given text.
- $\mathbf{y} \in \mathbf{Y} = \{\mathbf{y_1}, \mathbf{y_2}, ..., \mathbf{y_k}\}$ be the sentiment label, where $\mathbf{k}$ is the number of sentiment classes.
  - $\mathbf{k} = \mathbf{3}$ for positive, negative, natural

Objective

To learn a function $\mathbf{f}$, such that:

$$\mathbf{f} : \mathbf{X} \rightarrow \mathbf{y}$$

Where $\mathbf{f}$ is typically a machine learning model, such as Naive Bayes, SVM, neural networks, or BERT.

Probability-Based Representation

In probabilistic terms, sentiment analysis aims to estimate:

$$\hat{\mathbf{y}} = \mathbf{argmax}_{\mathbf{y} \in \mathbf{Y}} \mathbf{P}(\mathbf{y}|\mathbf{X})$$

$\mathbf{P}(\mathbf{y}|\mathbf{X})$ is the probability of sentiment $\mathbf{y}$ given the text $\mathbf{X}$. It can be obtained by the following probability models.

- Naive Bayes: $\mathbf{P}(\mathbf{y}|\mathbf{X}) \approx \mathbf{P}(\mathbf{X}|\mathbf{y})\mathbf{P}(\mathbf{y})$.
- Neural Networks: $\mathbf{P}(\mathbf{y}|\mathbf{X}) = \mathbf{softmax}(\mathbf{W} \cdot \mathbf{X} + \mathbf{b})$, where $\mathbf{W}$ and $\mathbf{b}$ are learned parameters.
- Transformer models (BERT): Use attention mechanisms to learn contextual embeddings and predict $\mathbf{P}(\mathbf{y}|\mathbf{X})$.

Python Library of Probability Models

- Naive Bayes – Scikit-learn
- Neural Networks – Keras/TensorFlow
- BERT – Hugging Face

Model	Description	Key Python Library
Naive Bayes	A simple probabilistic model using **TF-IDF** features for text classification.	`scikit-learn`
Neural Networks	A feedforward network for binary sentiment classification. Uses **TF-IDF** features.	`Keras/TensorFlow`
BERT (Transformers)	A state-of-the-art transformer model that handles sentiment analysis with pre-trained weights.	`transformers` (Hugging Face)

Note, instead of **TF/IDF** features, you can use word embeddings (GloVe, Wrod2Vec).

Generating Response

Once the emotion $\hat{\mathbf{y}}$ is detected, the next step is to generate a response $\mathbf{r}$ conditioned on the detected emotion.

Definiton

$$\mathbf{r} = \mathbf{g}(\hat{\mathbf{y}}, \mathbf{X})$$

Note, $\mathbf{g}(\cdot)$ can be

- Rule-based systems: For predefined responses.
- Retrieval-based systems: For selecting a response from a database of responses.
- Generative models (e.g., GPT, LSTM): For generating a response dynamically.

Rule-based Response Systems

Let: $\mathbf{X}$ be a input sequence, $\hat{\mathbf{y}}$ is detected emotion, $\mathbf{f_{rules}}$ is the set of predefined responses for $\hat{\mathbf{y}}$.

Definition

A rule-based response generation system uses predefined mappings between emotions and responses.

$$\mathbf{r}(\hat{\mathbf{y}}) = \mathbf{f_{rules}}(\mathbf{X})$$

For example:

- $\mathbf{X} =$ **I feel so happy today**.
- $\hat{\mathbf{y}} =$ **positive**
- $\mathbf{r} =$ **This is great to hear! Keep up the positivity!**

Retrieval-based Response Systems

Definition

- The system retrieves a response from a database based on the detected emotion.
- Let $\mathbf{X}$ be an input sentence.
- Let $\mathbf{R}_{\hat{\mathbf{y}}}$ the set of all possible responses for a detected emotion $\hat{\mathbf{y}}$.
- The system retrieves the best matching response:

$$\mathbf{r}(\hat{\mathbf{y}}) = \mathbf{argmax}_{\mathbf{r} \in \mathbf{R}_{\hat{\mathbf{y}}}} \ \mathbf{similarity}(\mathbf{r}, \mathbf{X})$$

$$\mathbf{similarity}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}|| * ||\mathbf{b}||}$$

- Note, $\mathbf{a}$ and $\mathbf{b}$ are vectors (embeddings, obtained from GloVe or Word2Vec).

An Example of Retrieval-based Response

- User: "I'm feeling sad."
- Responses might be:
  - "I'm sorry to hear that. Do you want to talk about it?"
  - "That sounds really tough. I'm here for you."
  - "Is there anything I can do to help you feel better?"
  - "I understand. Sometimes, talking to someone can help."
  - "It's okay to feel this way. You're not alone."
  - "Would you like to listen to some relaxing music?"
  - "Maybe a short walk or deep breathing could help. Want to try?"
- The best response (with the highest similarity score) will be:
  - "I'm sorry to hear that. Do you want to talk about it?"

Generative Model Response Systems (GPT-3, LSTM)

Defintion

In a generative model, the response is generated word by word, conditioned on both the input text $\mathbf{X}$ and the detected emotion $\hat{\mathbf{y}}$.

$$\mathbf{P}(\mathbf{r}|\mathbf{X}, \hat{\mathbf{y}}) = \prod_{\mathbf{t}=1}^{\mathbf{T}} \mathbf{P}(\mathbf{r_t}|\mathbf{r_1}, \mathbf{r_2}, ..., \mathbf{r_{t-1}}, \mathbf{X}, \hat{\mathbf{y}})$$

- $\mathbf{r_t}$ is the $\mathbf{t^{th}}$ word of the response, and $\mathbf{T}$ is the length of the response.
- The language model can be learned by RRN with LSTM.
- The language model can be learned by the transformers (GPT, BERT)

An Example of Generative models Response

- User: "I'm feeling sad."
- GPT-Based Response (Transformer Model)
  - "I'm sorry to hear that. Do you want to talk about it? Sometimes sharing your feelings can help."
- LSTM-Based Response (RNN Model)
  - "That sounds really tough. Remember, you're not alone, and things will get better."

Sentiment Analysis

Data Samples.

ID	Input Sentence	Sentiment Label
1	"I love this product! It's amazing!"	**Positive**
2	"This is the worst experience I've ever had."	**Negative**
3	"The movie was okay, nothing special."	**Neutral**
4	"I'm so excited about my vacation next week!"	**Positive**
5	"I feel really sad and lonely today."	**Negative**
6	"The delivery was delayed, but the product is good."	**Neutral**
7	"Absolutely fantastic service! Highly recommended."	**Positive**
8	"I'm disappointed with the quality of this item."	**Negative**
9	"It is what it is, not much to say."	**Neutral**
10	"This meal is delicious! I love the flavors."	**Positive**

Sentiment Analysis

Embedding vectors

ID	Input Sentence	10-Dimensional Sentence Embedding	Class Label
1	"I love this product! It's amazing!"	[0.28, 0.14, -0.21, 0.33, -0.05, 0.20, 0.12, 0.35, -0.12, 0.07]	Positive
2	"This is the worst experience I've ever had."	[-0.34, 0.11, -0.45, 0.27, -0.19, -0.30, 0.05, 0.12, -0.27, -0.15]	Negative
3	"The movie was okay, nothing special."	[0.06, -0.05, -0.17, 0.15, -0.02, 0.10, 0.08, 0.22, -0.08, 0.10]	Neutral
4	"I'm so excited about my vacation next week!"	[0.32, 0.28, -0.12, 0.40, -0.10, 0.22, 0.18, 0.41, -0.14, 0.15]	Positive
5	"I feel really sad and lonely today."	[-0.40, 0.05, -0.39, 0.12, -0.25, -0.20, -0.14, -0.10, -0.25, -0.22]	Negative
6	"The delivery was delayed, but the product is good."	[0.12, 0.01, -0.14, 0.20, -0.06, 0.08, 0.11, 0.18, -0.19, 0.09]	Neutral
7	"Absolutely fantastic service! Highly recommended."	[0.41, 0.30, -0.10, 0.45, -0.04, 0.25, 0.19, 0.50, -0.08, 0.20]	Positive
8	"I'm disappointed with the quality of this item."	[-0.38, 0.08, -0.42, 0.14, -0.18, -0.28, -0.07, 0.05, -0.21, -0.18]	Negative
9	"It is what it is, not much to say."	[0.02, -0.04, -0.11, 0.10, -0.01, 0.06, 0.05, 0.15, -0.06, 0.05]	Neutral
10	"This meal is delicious! I love the flavors."	[0.37, 0.25, -0.09, 0.38, -0.03, 0.23, 0.17, 0.42, -0.10, 0.18]	Positive

Sentiment Analysis

Data instances, where $\mathbf{Y} = \{\textbf{negative}, \textbf{natural}, \textbf{positive}\} = \{\mathbf{0}, \mathbf{1}, \mathbf{2}\}$

ID	10-Dimensional Sentence Embedding	Class Label
1	[0.28, 0.14, -0.21, 0.33, -0.05, 0.20, 0.12, 0.35, -0.12, 0.07]	2
2	[-0.34, 0.11, -0.45, 0.27, -0.19, -0.30, 0.05, 0.12, -0.27, -0.15]	0
3	[0.06, -0.05, -0.17, 0.15, -0.02, 0.10, 0.08, 0.22, -0.08, 0.10]	1
4	[0.32, 0.28, -0.12, 0.40, -0.10, 0.22, 0.18, 0.41, -0.14, 0.15]	2
5	[-0.40, 0.05, -0.39, 0.12, -0.25, -0.20, -0.14, -0.10, -0.25, -0.22]	0
6	[0.12, 0.01, -0.14, 0.20, -0.06, 0.08, 0.11, 0.18, -0.19, 0.09]	1
7	[0.41, 0.30, -0.10, 0.45, -0.04, 0.25, 0.19, 0.50, -0.08, 0.20]	2
8	[-0.38, 0.08, -0.42, 0.14, -0.18, -0.28, -0.07, 0.05, -0.21, -0.18]	0
9	[0.02, -0.04, -0.11, 0.10, -0.01, 0.06, 0.05, 0.15, -0.06, 0.05]	1
10	[0.37, 0.25, -0.09, 0.38, -0.03, 0.23, 0.17, 0.42, -0.10, 0.18]	2

Main Categories of Machine Learning

1. Supervised Learning
   - The model is trained on labeled data, where the input-output pairs are provided.
2. Unsupervised Learning
   - The model is trained on unlabeled data to find hidden structures or patterns.
3. Reinforcement Learning
   - The model learns by interacting with an environment and receiving rewards or penalties.

Supervised Learning

Definition

In supervised learning, the model is trained on a labeled dataset, where both the input (features) and the corresponding output (label or target) are provided.

Goal

The goal is for the model to learn the mapping between inputs and outputs so that it can predict the output for new, unseen inputs.

An Example

Sentiment analysis is a supervised learning task in natural language processing (NLP) where the goal is to predict the sentiment or emotional tone of a given text, such as whether a review, tweet, or piece of content expresses positive, negative, or neutral feelings.

Algorithms (Models) on Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Neural Networks
- Gradient Boosting Machines (GBM)
- AdaBoost
- Naive Bayes

Work Flow on Supervised Learning

Problem Definition

- Understand the task: Determine whether it's a classification (predicting categories) or regression (predicting continuous values) problem.
- Set objectives: Clarify what you want to achieve.

Data Collection

- Gather labeled data: Supervised learning requires a dataset with input-output pairs, where the input features and the corresponding output (target or label) are provided.
- Sources of data: Collect data from various sources like databases, CSV files, web scraping, APIs, or sensors.

Work Flow on Supervised Learning

Data Preprocessing

- Data Cleaning: Handle missing values, outliers, and inconsistencies. For example, filling missing data with mean/median, removing or imputing missing labels.
- Feature Selection: Identify relevant features (variables) that help in making predictions. Remove irrelevant or redundant features to improve model performance.
- Feature Engineering: Create new features from existing data that may help improve the model. For instance, creating interaction terms, normalizing or scaling data.
- Encoding Categorical Variables: If features are categorical (like text or labels), convert them into numerical form using techniques like one-hot encoding, label encoding, etc.

Work Flow on Supervised Learning

Data Splitting

- Train-Test Split: Divide the dataset into two subsets: one for training the model and another for testing. The typical ratio is 70-80% for training and 20-30% for testing.
- Cross-validation (optional): For more reliable evaluation, use techniques like k-fold cross-validation to split the data into multiple subsets and validate the model on each.

Model Selection

- Choose a model: Depending on the problem, choose a suitable supervised learning algorithm.
- Consider complexity: Select simpler models) for simple tasks, and more complex models for complex tasks with large datasets.

Work Flow on Supervised Learning

Model Training

- Fit the model to the training data: Use the training dataset to teach the model how to make predictions.
- During this step, the model learns the relationship between input features and target values.

Model Evaluation

- Test the model: Use the test dataset to assess how well the model generalizes to unseen data.
- Performance Metrics:
  - Classification: Accuracy, Precision, Recall, F1-score, AUC-ROC, Confusion Matrix, binary cross entropy, multi-class cross entropy
  - Regression: Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared.

Work Flow on Supervised Learning

Model Deployment

- Deploy the model: After achieving satisfactory performance, deploy the model for real-time predictions or batch predictions.
- Integration: Integrate the model into an application or system where it can make predictions based on new data.

Evaluation Metrics for Classification

- Let $\mathbf{n}$ be the size of data samples of testing data.
- Let $\mathbf{y^i}$ be the true class label of $\mathbf{i^{th}}$ data instance in the test data set.
- Let $\mathbf{\hat{y}^i}$ be the assigned class label of $\mathbf{i^{th}}$ data instance in the test data set by our trained model.

Accurracy

$$\mathbf{ACC}_{\mathcal{D}_{\mathbf{test}}} = \frac{\mathbf{1}}{\mathbf{n}} \sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{L}(\mathbf{\hat{y}^i}, \mathbf{y^i})$$

where:

$$\mathbf{L}(\mathbf{\hat{y}^i}, \mathbf{y^i}) = \begin{cases} 1 & \text{if } \mathbf{\hat{y}^i} = \mathbf{y^i} \\ 0 & \text{if } \mathbf{\hat{y}^i} \neq \mathbf{y^i} \end{cases}$$

An Accuracy Example

y_ass	y_true
-1	-1
-1	-1
-1	-1
-1	-1
-1	-1
-1	1
-1	-1
-1	-1
-1	-1
-1	-1
-1	1
-1	-1
-1	-1
-1	-1
-1	-1
-1	1
-1	1
-1	-1
-1	1
-1	1
-1	-1
-1	-1
-1	-1
-1	-1
-1	-1
-1	-1
-1	-1

Figure: Assigned labels vs. true labels

- $\mathbf{Acc} = \frac{21}{27} = \mathbf{0.78}$

Contingency Table

- Let $\mathbf{C} = \{-\mathbf{1}, \mathbf{1}\}$ be the true class label and $\mathbf{A} = \{-\mathbf{1}, \mathbf{1}\}$ be the assigned class label. The contingency table is defined as

A \ C	C = 1	C = -1
A=1	Count(C=1, A=1) α	Count(C=-1, A=1) β
A=-1	Count(C=1, A=-1) γ	Count(C=-1, A=-1) δ

Recall, Precision, F1-Score

Recall Definition

$$\textbf{recall} = \frac{\alpha}{\alpha + \gamma}$$

Precision Definitiion

$$\textbf{precision} = \frac{\alpha}{\alpha + \beta}$$

F1-Score

$$\textbf{F1} - \textbf{Score} = 2 * \frac{\textbf{Precision} \times \textbf{Recall}}{\textbf{Precision} + \textbf{Recall}}$$

An Example

y_true	y_ass
-1	-1
1	1
-1	-1
1	-1
1	-1
1	1
1	-1
-1	1
1	1
-1	-1
1	1
1	-1
-1	1
1	-1
-1	-1
1	1
-1	1
1	-1
-1	1
1	1
-1	-1
-1	-1
1	-1
1	-1
-1	1
-1	-1
1	1

Determine:

- The contingency table
- Precision
- Recall
- F1-Score

Summary

- Affective Computing
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